

1) Find the rank of the matrix $\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix}$

Ans)

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix} \quad R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} -1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 4 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 - 4 \times R_2$$

$$\begin{bmatrix} -1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \therefore \text{Rank}(A) = 2$$

2) Find the Eigen values of the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ what are the Eigen values of A^2, A^{-1} without using its characteristic equation

Ans)

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$$

Eigen Value of triangular matrix are diagonal element of that matrix

$\therefore$ Eigen Value are 1, 3

$$A^2 = 1^2, 3^2 \\ = 1, 9$$

$$A^{-1} = \frac{1}{1}, \frac{1}{3}$$

11) Solve the following linear system of equations using Gauss elimination method

$$x + 2y - z = 3$$

$$3x - y + 2z = 1$$

$$2x - 2y + 3z = 2$$

Ans a.

$$\tilde{A} = [A \cdot B] = \begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 3 & -1 & 2 & : & 1 \\ 2 & -2 & 3 & : & 2 \end{bmatrix} \quad R_2 \rightarrow R_2 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 0 & -7 & 5 & : & -8 \\ 2 & -2 & 3 & : & 2 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 0 & -7 & 5 & : & -8 \\ 0 & -6 & 5 & : & -4 \end{bmatrix} \quad R_3 \rightarrow R_3 - \frac{-6}{-7} R_2$$

$$\begin{bmatrix} 1 & 2 & -1 & : & 3 \\ 0 & -7 & 5 & : & -8 \\ 0 & 0 & \frac{5}{7} & : & \frac{20}{7} \end{bmatrix}$$

$$\frac{5}{7} z = \frac{20}{7}$$

$$z = \underline{\underline{4}}$$

$$-7y + 5z = -8$$

$$-7y + 20 = -8$$

$$-7y = -28$$

$$y = \underline{\underline{4}}$$

$$x + 2y - z = 3$$

$$x + 8 - 4 = 3$$

$$x + 4 = 3$$

$$x = \underline{\underline{-1}}$$

11) b) Find the eigen values and eigen vectors of $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

Ans. b)

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

$$\lambda^3 - \text{tr}(A)\lambda^2 + (m_{11} + m_{22} + m_{33})\lambda - |A| = 0$$

$$|A| = \begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{vmatrix} = 0$$

$$\lambda^3 - 18\lambda^2 + (5 + 20 + 20)\lambda - 0 = 0$$

$$\lambda^3 - 18\lambda^2 + 45\lambda - 0 = 0$$

$$m_{11} = \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} = 5$$

$$\lambda = 15, 3, 0$$

When $\lambda = 15$

$$m_{22} = \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} = 20$$

$$(A - \lambda I)x = 0$$

$$(A - 15I)x = 0$$

$$m_{33} = \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix} = 20$$

$$\text{tr}(A) = 8 + 7 + 3 = 18$$

$$\begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 - \left(\frac{6}{-7}\right)R_1$$

$$\begin{bmatrix} -7 & -6 & 2 \\ 0 & \frac{-20}{7} & \frac{-40}{7} \\ 2 & -4 & -12 \end{bmatrix} \quad R_3 \rightarrow R_3 - \frac{2}{-7}R_1$$

$$\begin{bmatrix} -7 & -6 & 2 \\ 0 & \frac{-20}{7} & \frac{-40}{7} \\ 0 & \frac{-40}{7} & \frac{-80}{7} \end{bmatrix} \quad R_3 \rightarrow R_3 - \frac{\left(\frac{-40}{7}\right)}{\left(\frac{-20}{7}\right)}R_2$$

11 Ans. b)

$$\begin{bmatrix} -7 & -6 & 2 \\ 0 & -20/7 & -40/7 \\ 0 & 0 & 0 \end{bmatrix}$$

Let $x_3 = 1$

$$-\frac{20}{7}x_2 + \frac{-40}{7}x_1 = 0$$

$$-\frac{20}{7}x_2 + \frac{40}{7} = 0$$

$$x_2 = \underline{\underline{-2}}$$

$$-7x_1 - 6x_2 + 2x_3 = 0$$

$$-7x_1 + 12 + 2 = 0$$

$$-7x_1 + 14 = 0$$

$$x_1 = \underline{\underline{2}}$$

$$X_1 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

When $\lambda = 3$

$$(A - \lambda I)x = 0$$

$$(A - 3I)x = 0$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 - \left(\frac{-6}{5}\right)R_1$$

$$\begin{bmatrix} 5 & -6 & 2 \\ 0 & \frac{-16}{5} & \frac{-8}{5} \\ 2 & -4 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{2}{5}R_1$$

$$\begin{bmatrix} 5 & -6 & 2 \\ 0 & \frac{-16}{5} & \frac{-8}{5} \\ 0 & \frac{-8}{5} & \frac{-4}{5} \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{\left(\frac{-8}{5}\right)}{\left(\frac{-16}{5}\right)}R_2$$

Ans (1b)

$$\begin{bmatrix} 5 & -6 & 2 \\ 0 & \frac{-16}{5} & \frac{-8}{5} \\ 0 & 0 & 0 \end{bmatrix}$$

Let $x_3 = 1$

$$-\frac{16}{5}x_2 - \frac{8}{5}x_1 = 0$$

$$-\frac{16}{5}x_2 - \frac{8}{5} = 0$$

$$x_2 = \underline{\underline{-\frac{1}{2}}}$$

$$5x_1 - 6x_2 + 2x_3 = 0$$

$$5x_1 + 3 + 2 = 0$$

$$5x_1 + 5 = 0$$

$$x_1 = \underline{\underline{-1}}$$

$$X_2 = \begin{bmatrix} -1 \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

When $\lambda = 0$

$$(A - \lambda I)x = 0$$

$$(A - 0I)x = 0$$

$$(A)x = 0$$

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad R_2 \rightarrow R_2 - \left(\frac{-6}{8}\right)R_1$$

$$\begin{bmatrix} 8 & -6 & 2 \\ 0 & \frac{5}{2} & -\frac{5}{2} \\ 2 & -4 & 3 \end{bmatrix} \quad R_3 \rightarrow R_3 - \left(\frac{2}{8}\right)R_1$$

$$\begin{bmatrix} 8 & -6 & 2 \\ 0 & \frac{5}{2} & -\frac{5}{2} \\ 0 & -\frac{5}{2} & \frac{5}{2} \end{bmatrix} \quad R_3 \rightarrow R_3 + R_2$$

Ans)

Ans 11)

$$\begin{bmatrix} 8 & -6 & 2 \\ 0 & 5/2 & -5/2 \\ 0 & 0 & 0 \end{bmatrix}$$

Let $x_3 = 1$

$$\frac{5}{2}x_2 - \frac{5}{2}x_3 = 0$$

$$\frac{5}{2}x_2 - \frac{5}{2}$$

$$x_2 = \frac{1}{1}$$

$$8x_1 - 6x_2 + 2x_3 = 0$$

$$8x_1 - 6 + 2 = 0$$

$$8x_1 - 4 = 0$$

$$x_1 = \frac{4}{8} = \frac{1}{2}$$

$$x_3 = \begin{bmatrix} \frac{1}{2} \\ 1 \\ 1 \end{bmatrix}$$

12)

Solve the following linear system of equations using Gauss elimination method

$$2x - 2y + 4z = 0$$

$$-3x + 3y - 6z + 5w = 15$$

$$x - y + 2z = 0$$

Ans) 12a)

$$\tilde{A} = [A \cdot B] = \left[\begin{array}{cccc|c} 2 & -2 & 4 & 0 & 0 \\ -3 & 3 & -6 & 5 & 15 \\ 1 & -1 & 2 & 0 & 0 \end{array} \right] \quad R_3 \leftrightarrow R_1$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & 0 & 0 \\ -3 & 3 & -6 & 5 & 15 \\ 2 & -2 & 4 & 0 & 0 \end{array} \right] \quad \begin{array}{l} R_2 \rightarrow R_2 + 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 5 & 15 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Ans)
12a

$$\text{Rank}(A) = 2$$

$$\text{Rank}(A^T) = 2$$

$$\text{no of unknowns} = 4$$

$$x - y - 2z = 0$$

$$5w = 15$$

$$w = 5$$

$$\text{Let } y = t_1, z = t_2$$

then

$$x = t_1 + 2t_2$$

12b

Find the matrix of transformation that diagonalize the matrix $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$
also write the diagonal matrix

Ans)
12b

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

$$\text{tr}(A) = -7$$

$$|A| = 6$$

$$\lambda^2 + 7\lambda + 6 = 0$$

$$\lambda = -1, -6$$

$$\text{When } \lambda = -1$$

$$(A + I)x = 0$$

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad R_2 \rightarrow R_2 + \frac{1}{2}R_1$$

$$\begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix}$$

$$\text{Let } x_2 = 1$$

$$-4x_1 + 2x_2 = 0$$

$$-4x_1 + 2 = 0$$

$$x_1 = \frac{1}{2}$$

$$x_1 = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$$

Ans

12b

$$\lambda = -6$$

$$(A + 6I)x = 0$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$x_1 + 2x_2 = 0 \quad \text{Let } x_2 = 1$$

$$x_1 + 2 = 0$$

$$x_1 = -2$$

$$x_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$D = P^{-1}AP$$

$$P = \begin{bmatrix} \frac{1}{2} & -2 \\ 1 & 1 \end{bmatrix} \quad P^{-1} = \begin{bmatrix} \frac{2}{5} & \frac{4}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix} \quad A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

$$D = \begin{bmatrix} \frac{2}{5} & \frac{4}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix} \times \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \times \begin{bmatrix} \frac{1}{2} & -2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -6 \end{bmatrix}$$